LU DECOMPOSITION

LU without row swaps

For an invertible A, the plain factorization exists exactly when no pivot is zero.

PLAIN LU

$$A = LU$$

HOLDS WHEN — every leading principal submatrix of A is nonsingular: the upper-left $k \times k$ block has nonzero determinant for $k = 1, \ldots, n$

✓ Factorization exists

All leading principal minors nonzero

Elimination never meets a zero in a pivot position.

BOUNDARY

✗ No plain LU

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

The $(1,1)$ entry is zero, and no multiple of row 1 can create a nonzero pivot.

A zero in any pivot position during elimination

No row-addition operation can produce a nonzero pivot; a row swap is required.

THE FIX — $PA = LU$ After swapping the two rows, elimination proceeds immediately: the pivoted version is needed.